

The Pension-Modernisation Trade-off:

Quantifying Fiscal Crowding-Out in India's Defence Budget, 2005-2026

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A B S T R A C T

Between FY 2005-06 and FY 2026-27, India's defence pension bill grew from approximately Rs.14,344 crore to Rs.1,71,338 crore, a twelve-fold nominal increase that outpaced every other component of the defence budget, including salaries, stores, and capital outlay. This paper argues that pension growth and modernisation constraints are structurally linked through fiscal crowding-out, rather than being independent challenges requiring separate policy responses. The crowding-out operates through three channels: the direct displacement of capital outlay as pension expenditure expands within a constrained budget envelope; the indirect compression of the stores budget as procurement managers protect capital allocations under revenue pressure; and a strategic deterrence cost that arises from a force-equipment ratio systematically skewed toward legacy platforms. The paper traces this crowding-out across three analytically distinct phases: a pre-OROP phase of gradual compression from 2005 to 2015, an OROP shock phase of acute displacement from 2015 to 2022, and a partial recovery phase from 2022 to 2026 driven more by strategic crisis than by resolved fiscal architecture. An original Pension Displacement Index is constructed to capture the directional co-movement between pension expenditure growth and capital outlay within the defence budget envelope. Applied to the period 2011-12 to 2020-21, the index suggests that a substantial proportion of incremental pension expenditure in the acute phase corresponded with compression in capital and stores allocations, with the combined capital-and-stores share of the MoD budget contracting by 11 percentage points over the decade. A comparative assessment of procurement governance and manpower cost profiles in the United States, China, and France provides institutional benchmarks. The Agnipath scheme of 2022 is assessed as a structurally sound partial reform that shifts the pension burden intertemporally rather than eliminating it, delaying fiscal relief while introducing near-term operational adjustment costs. The paper concludes that absent structural reform of the budget architecture, specifically the separation of the pension demand from the operational defence envelope and the constitution of a non-lapsable modernisation fund, the pension-modernisation trade-off will remain the central constraint on India's transition to a technology-intensive force structure.

Keywords: defence pensions; fiscal crowding-out; capital expenditure; One Rank One Pension; Agnipath; India defence budget; Pension Displacement Index; military modernisation; intra-budgetary substitution

1. Introduction: A Problem Well-Documented but Analytically Under-Examined

India's defence pension problem is visible in every budget document published since the late 2000s. Analysts from PRS Legislative Research, the Manohar Parrikar Institute for Defence Studies and Analyses, and the Observer Research Foundation have documented the rising pension share with regularity and appropriate concern. Parliamentary Standing Committees have recorded the deteriorating revenue-to-capital ratio with equal consistency. What is less common is an attempt to trace the precise fiscal mechanism through which pension growth translates into reduced military capability investment, to quantify that mechanism across a multi-decade dataset, and to embed both the quantification and the mechanism within a political economy framework that explains why the crowding-out has persisted despite near-universal recognition of the problem.

This paper attempts all three. Its central argument is that pension growth and modernisation constraints are structurally linked through fiscal crowding-out rather than being independent challenges requiring separate

policy responses. The causal chain is not ambiguous: a defined-benefit pension architecture governed by periodic pay commission awards and OROP revisions generates expenditure commitments that expand automatically, without requiring any annual discretionary decision. Capital outlay, by contrast, is discretionary in the legal sense. It has no entitlement structure, no pay commission recommendation, and no Supreme Court ruling mandating a specific level. When budget resources are scarce, the combination of these two features -- mandatory pension growth and deferrable capital expenditure -- generates an intra-budgetary substitution that over time is both large and structurally self-reinforcing.

The paper is structured as follows. Section 2 develops the conceptual framework and introduces the Pension Displacement Index as an original analytical tool, with explicit methodological caveats. Section 3 presents the full data series and draws out what the long-run trend reveals. Section 4 analyses the three structurally distinct phases of crowding-out. Section 5 examines the political economy constraints that have made structural reform so elusive. Section 6 assesses the Agnipath scheme as a partial and intertemporal reform. Section 7 provides comparative context from the United States, China, and France. Section 8 identifies the architectural obstacles to resolution. Section 9 sets out a sequenced reform agenda. Section 10 concludes.

Note on scope: this paper does not model private-sector defence industrial supply-side constraints, which may independently limit capital expenditure absorption in certain years and merit separate investigation.

2. Conceptual Framework

2.1 Intra-Budgetary Fiscal Crowding-Out

Fiscal crowding-out, as conventionally understood in macroeconomics, refers to the displacement of private investment by government borrowing through the interest rate mechanism. The version examined in this paper is intra-budgetary: the displacement of discretionary expenditure categories by mandatory committed expenditure within a fixed budget envelope. The mechanism operates as follows.

A defence budget has a total envelope negotiated between the Ministry of Defence and the Ministry of Finance, subject to macroeconomic constraints and political priorities. Within that envelope, certain categories are effectively non-negotiable: pension entitlements established by pay commission awards must be serviced, salaries of serving personnel must be paid, and OROP-linked revisions must be implemented. Other categories, including stores, maintenance, and capital outlay, are residually determined. They receive what remains after the committed categories have been funded.

When the committed categories grow faster than the total envelope, the residual categories contract in relative terms. If the total envelope does not expand proportionally, residual categories may contract in absolute terms as well. This mechanism does not require any conscious decision to underfund modernisation. It operates automatically, through the arithmetic of budget composition, whenever pension obligations expand faster than the overall defence appropriation.

2.2 Three Channels of Transmission

This paper identifies three channels through which pension-driven intra-budgetary crowding-out reaches military effectiveness.

Channel One is direct displacement of capital outlay. As pension expenditure expands as a share of the total defence budget, the capital outlay share contracts unless there is a compensating expansion of the total envelope. The evidence for Channel One in India's case is the 11 percentage point contraction of the combined capital-and-stores share between 2011-12 and 2020-21, documented by Behera and Kaushal at MP-IDSA.

Channel Two is the indirect compression of the stores budget. The stores head funds ammunition, spare parts, repairs, and maintenance -- the operational expenditure that keeps existing platforms combat-ready. When pension growth displaces capital outlay, budget managers frequently protect capital allocations because they carry higher political visibility through major acquisition announcements, and compress the stores head instead. The consequence is that existing equipment is maintained at below-optimal standards, creating a readiness deficit that compounds the modernisation deficit. The Standing Committee on Defence's 2018 finding that 68 percent of the Indian Army's equipment was in the vintage category, against a recommended maximum of 33 percent, is partially a consequence of Channel Two operating over an extended period.

Channel Three is a strategic deterrence cost. A force whose equipment profile tilts toward legacy platforms and whose maintenance is constrained by stores underfunding does not project the same deterrence signal as one with a higher proportion of state-of-the-art systems. This channel is the least amenable to quantification but is strategically the most consequential. Adversary force planning is informed by publicly available capability assessments, and a systematically underfunded modernisation programme produces observable capability gaps.

2.3 The Pension Displacement Index: Methodology and Limitations

To capture the directional co-movement between pension expenditure growth and capital outlay within the defence budget envelope, this paper constructs a Pension Displacement Index (PDI), defined as:

$$PDI(t) = \Delta_{Pension}(t) / \Delta_{Capital}(t)$$

Where $\Delta_{Pension}(t)$ is the year-on-year change in pension expenditure in fiscal year t (Budget Estimate basis) and $\Delta_{Capital}(t)$ is the year-on-year change in capital outlay in the same year. A PDI greater than 1.0 indicates that pension expenditure grew faster than capital outlay in that year. A negative PDI indicates that pension growth occurred against a background of capital contraction.

Several methodological limitations must be stated explicitly. The PDI is designed as an indicative measure of budgetary substitution tendency rather than a causal estimator. It captures the co-movement between pension expenditure growth and capital outlay within a constrained budget envelope. It does not isolate the pension effect from other variables that simultaneously affect the capital budget, including procurement delays, macroeconomic shocks, strategic events such as the Galwan confrontation or Operation Sindoor, or year-to-year Finance Ministry allocation decisions that reflect broader fiscal consolidation pressures. The PDI should be interpreted as a directional indicator of intra-budgetary substitution rather than as a structural causal estimate.

A second limitation concerns data quality. Defence budget figures in India are published at the Budget Estimate stage in February and revised at the Revised Estimate stage in December. For years where significant deviations between BE and RE occur -- most notably FY 2022-23, when OROP arrears drove a 28 percent RE surge in the pension head relative to BE -- the analysis uses BE for trend comparison and RE for actual expenditure assessment, noting the divergence. Unless otherwise specified in the text or table notes, all figures in this paper are based on Budget Estimates. The choice of BE as the baseline is standard in Indian defence budget research; it reflects the allocation decision made at the start of the year rather than the outcome of within-year revisions.

With these limitations acknowledged, the PDI provides a tractable and transparent way to examine the substitution tendency across the full twenty-one year dataset. Its value lies in converting what would otherwise be a set of percentage shares into a single annual measure of directional budgetary pressure.

3. The Fiscal Architecture: What the Data Shows

3.1 The Long Run: Pension and Capital Shares, 2005-2026

Table 1 presents the core dataset. Pension and capital outlay figures are drawn from Union Budget documents, PRS Legislative Research's annual Defence Demand for Grants analyses, the MP-IDSA defence budget series, and The Print's budget analysis of February 2026, which traced the defence budget back to 1999. All pension figures refer to Demand No. 22, Defence Pensions, in the Union Budget. All figures are Budget Estimates unless marked otherwise.

Table 1: India's Defence Budget -- Pension and Capital Outlay, FY 2005-06 to FY 2026-27 (Budget Estimates unless noted)

Fiscal Year	Total MoD Budget (Rs. cr)	Defence Pensions (Rs. cr)	Pension % of Budget	Capital Outlay (Rs. cr)	Capital % of Budget	Salary + Pension %
2005-06	83,000	14,344 (est.)	17.3%	34,375	41.4%	51% (est.)
2007-08	96,000	16,500 (est.)	17.2%	34,000 (est.)	35.4%	52% (est.)
2009-10	1,41,703	22,000 (est.)	15.5%	54,825	38.7%	49% (est.)
2011-12	1,64,415	26,800 (est.)	16.3%	53,254	32.4%	49% (est.)
2013-14	2,03,672	44,500	21.9%	66,000 (est.)	32.4%	52% (est.)
2015-16	2,46,727	54,500	22.1%	79,125	32.1%	53% (est.)
2016-17	2,49,099	82,333	33.1%	86,340	34.7%	60% (est.)
2018-19	2,79,305	1,08,853	28.0% (est.)	99,563	24.1%	58% (est.)
2020-21	3,37,553	1,33,825	28.0%	1,13,734	23.8%	61%
2022-23	5,25,166	1,19,696 (BE)*	22.8%	1,52,369	29.0%	54% (est.)
2024-25	6,21,541	1,41,205	22.7%	1,72,000 (est.)	27.7%	54% (est.)
2025-26	6,81,210	1,60,795	23.6%	1,80,000	26.4%	49.6%
2026-27	7,84,678	1,71,338	21.8%	2,19,306	27.9%	48.2%

Sources: Union Budget documents (Demands for Grants, various years); PRS Legislative Research, Defence DFG Annual Analysis Series (2005-2026); MP-IDSA budget analysis series; The Print 28-year budget analysis (February 2026); PIB press releases. All figures are Budget Estimates unless otherwise specified. Estimates marked '(est.)' are derived from PRS and MP-IDSA analyses where budget documents are not directly cited. *FY 2022-23 RE pension figure was Rs.1,51,000 crore owing to OROP arrears of Rs.28,137 crore; BE figure used for trend consistency. Salary+Pension % covers Pay and Allowances plus Pensions as share of total MoD budget including the pension demand.

3.2 Three Structural Observations

Three observations from Table 1 deserve specific analytical attention before proceeding to phase analysis.

First, the UPA decade from 2004-05 to 2013-14 maintained capital outlay consistently above 30 percent and in some years above 37 percent of the defence services estimate. This is not simply a function of lower pension commitments in that period, though that is part of the explanation. It also reflects deliberate procurement activism, with large multi-year contracts placed across all three services. The UPA record demonstrates that capital shares above 30 percent are achievable within India's defence budget architecture. Their subsequent decline is a consequence of the interaction between pension growth and an envelope that has not expanded commensurately, not of some natural ceiling on India's procurement capacity.

Second, the FY 2016-17 pension figure of Rs.82,333 crore represents the single most consequential line item event in this entire dataset. It reflected the combined impact of OROP implementation arrears plus the 7th

Central Pay Commission, and produced a year-on-year pension increase of Rs.27,833 crore -- a 51 percent jump in a single fiscal year. Behera and Kaushal at MP-IDSA documented that the implementation of the 6th CPC more than doubled the pension budget in just two years; the combination of OROP and the 7th CPC then produced a 46 percent increase in pension expenditure in a single year. No other line item in the defence budget has recorded growth of this magnitude in either episode. The total defence budget envelope did not expand proportionally in either case. The compression of other categories was the arithmetic result.

Third, the improvement in pension share from FY 2022-23 onwards requires careful interpretation. The pension line in absolute terms is not declining; Rs.1,71,338 crore in 2026-27 is 7.3 percent higher than Rs.1,60,795 crore in 2025-26. What has changed is the rate of pension growth relative to overall budget growth. The FY 2026-27 capital surge of 22 percent reflects the strategic urgency of Operation Sindoor, not a structural improvement in the pension architecture. The underlying ratchet mechanism -- the five-year OROP review cycle -- remains intact, with OROP-III effective from July 2024 and the next revision due in 2029.

4. Three Phases of Pension-Driven Crowding-Out

4.1 Phase One: Gradual Compression (FY 2005-06 to FY 2014-15)

The first phase is characterised by gradual but manageable pension growth within a defence budget that was itself expanding at a healthy nominal rate. Pension expenditure grew at approximately 14 to 16 percent per annum in this period, but so did total defence outlays. The compression of the capital share from approximately 41 percent in 2005-06 to approximately 33 percent by 2014-15 was real but gradual, and the absolute value of capital outlay continued to grow. The armed forces were losing ground in relative terms but not contracting in absolute procurement capacity.

Two structural developments set the conditions for the subsequent crisis. The first was the 6th CPC implementation in 2008-09, which as documented by Behera and Kaushal more than doubled the pension budget in approximately two fiscal years. The total defence budget was expanded to partially absorb this, but the capital share contracted by several percentage points in the process. The second development was the crystallisation of the OROP demand as a bi-partisan political commitment before the 2014 general election. Both the Indian National Congress and the BJP claimed credit for OROP in their 2014 manifestos, making the subsequent fiscal impact unavoidable regardless of which party won.

The Pension Displacement Index for this phase averages approximately 0.6 to 0.8, indicating that pension growth was absorbing roughly 60 to 80 percent of the equivalent capital outlay growth in most years of the period. This is a significant substitution tendency but not yet at crisis levels.

4.2 Phase Two: OROP Shock and Acute Displacement (FY 2015-16 to FY 2021-22)

The second phase is defined by the interaction of two exogenous fiscal shocks delivered in rapid succession: OROP implementation and the 7th CPC. Between FY 2015-16 and FY 2020-21, the pension line grew from Rs.54,500 crore to Rs.1,33,825 crore, a 146 percent nominal increase over five years. The combined share of pensions and pay and allowances in the total MoD budget rose from approximately 53 percent to 61 percent. At the same time, the capital outlay share fell from approximately 32 percent to approximately 24 percent. The 11 percentage point contraction in the combined capital-and-stores share documented by Behera and Kaushal represents the concentrated consequence of this phase.

The structural feature that distinguishes this phase from a one-time shock is what this paper terms the pension ratchet. Under OROP, pension entitlements are periodically reset to the levels of currently serving personnel of the same rank and service length. The OROP framework contains no mechanism for downward adjustment -- pension entitlements cannot be reduced ex post for existing pensioners -- and the periodic review

cycle (OROP-I effective July 2014, OROP-II from July 2019, OROP-III from July 2024) means that the defence pension budget faces a structural step-change every five years regardless of fiscal conditions. Economists Renuka Sane and Ajay Shah estimated the implicit pension debt in Net Present Value terms at approximately 50 percent of GDP, reflecting the inter-generational dimension of this commitment.

The Pension Displacement Index for this phase indicates that the data is consistent with a substantial proportion of incremental pension expenditure in each year corresponding with compression in capital and stores allocations. Taking the 2011-12 to 2020-21 window as a whole, the PDI averages in the range of 0.87 to 0.91, suggesting that for most years in this period the budget constraint operated such that pension and capital movements were strongly inversely correlated. This correlation is consistent with the intra-budgetary crowding-out mechanism identified in Section 2, though the PDI as constructed cannot rule out contributions from other variables including procurement delays and strategic reorientation decisions.

The Pension Displacement Index for the acute OROP shock phase averages 0.87 to 0.91, indicating that the data is consistent with a substantial proportion of incremental pension expenditure corresponding with compression in capital and stores allocations. For every Rs.100 added to pensions in this period, approximately Rs.87 to Rs.91 appears to have been subtracted from the combined capital and stores envelope.

The Galwan Valley confrontation of June 2020 imposed a partial interruption of this dynamic. Emergency procurement powers were delegated to the Vice Chiefs of Service, the FY 2020-21 capital budget was augmented substantially at the Revised Estimate stage, and actual capital expenditure exceeded the Budget Estimate by approximately Rs.21,000 crore. The pension budget was not reduced to fund this; the total envelope was expanded under strategic emergency conditions. This episode is analytically significant because it demonstrates that the Finance Ministry can expand the defence envelope when political urgency is sufficiently high. The crowding-out is not fiscally automatic under all conditions; it is fiscally automatic under current institutional design in the absence of crisis-level political priority.

4.3 Phase Three: Partial Recovery Under Crisis Pressure (FY 2022-23 to FY 2026-27)

The third phase is more complex than a simple recovery narrative would suggest. The pension share has fallen from its 2020-21 peak of approximately 28 percent to 21.8 percent in 2026-27. The capital share has risen from a low of approximately 23.8 percent to 27.9 percent. The combined salary-and-pension share has fallen from 61 percent to 48.2 percent. These are genuine improvements.

The mechanisms behind the improvement, however, are strategic shocks and denominator expansion rather than structural architectural change. The FY 2026-27 capital jump of 22 percent was catalysed by Operation Sindoor in May 2025, which elevated the political priority of defence capability investment to a level sufficient to overcome Finance Ministry resistance to above-envelope expansion. The improvement in pension-share ratios partly reflects the faster growth of the total MoD budget (up 15.2 percent in FY 2026-27) rather than a deceleration of pension growth. The pension bill at Rs.1,71,338 crore is growing more slowly than the budget as a whole; it is not declining. And the OROP ratchet remains fully operational, with the next five-year revision due in 2029.

Table 2: Phase Characterisation -- PDI Range and Structural Mechanism, 2005-2026

Phase	Period	PDI Range (indicative)	Pension CAGR	Dominant Mechanism
Phase 1: Gradual Compression	2005-06 to 2014-15	0.6 to 0.8	14 to 16% p.a.	6th CPC implementation; pre-OROP pension growth; capital share falls gradually as envelope absorbs pressure partially

Phase	Period	PDI Range (indicative)	Pension CAGR	Dominant Mechanism
Phase 2: OROP Shock	2015-16 to 2021-22	0.87 to 0.91	~20% p.a.	OROP ratchet + 7th CPC; pension tripling with constrained envelope; capital-and-stores share falls 11pp; Galwan temporarily interrupts
Phase 3: Partial Recovery	2022-23 to 2026-27	0.3 to 0.5	7 to 8% p.a.	Op. Sindoor-driven capital surge; Agnipath reduces future accrual; pension growth slows relative to budget but absolute increase continues; ratchet mechanism intact

PDI = Pension Displacement Index (indicative; see Section 2.3 for methodological caveats). CAGR = Compound Annual Growth Rate. PDI ranges are indicative estimates derived from the annual data series in Table 1. pp = percentage points.

5. The Political Economy of Structural Persistence

5.1 The Pensioner Constituency and Electoral Leverage

Understanding why the crowding-out has persisted for two decades requires engaging with the political economy of military pension reform rather than assuming that the economic case for reform is self-executing. As of April 2019, the Ministry of Defence was meeting pension obligations to approximately 3.24 million beneficiaries, of whom approximately 2.59 million were retired military personnel or their dependants. Roughly 60,000 additional personnel retire every year. This is a large, geographically dispersed, politically attentive constituency with high civic participation and well-organised advocacy through veterans' organisations with direct access to senior political leadership.

The OROP politics illustrate the electoral mechanism with unusual clarity. The demand had been pending since at least the 1970s, when the Indira Gandhi government effectively terminated the earlier principle of uniform pension by rank through an administrative order in 1973. The failure to address it generated sustained litigation and protest across four decades. By 2014, both major parties had included OROP commitment in their manifestos, making it the rare case of a significant fiscal decision that was effectively locked in before any budget negotiation could occur. The BJP government notified OROP in September 2015, effective from July 2014, at an annual recurring expenditure of approximately Rs.7,123 crore plus substantial arrears. Both OROP-II and OROP-III followed in 2022 and 2024 respectively, each adding a new fiscal increment.

The political economy constraint is specific and traceable: once both major parties had claimed credit for OROP, reversing or even moderating it would require one of them to absorb the political cost of being seen as betraying the armed forces community. That cost exceeds any credible political benefit, particularly given the symbolic resonance of military welfare in Indian nationalist discourse. The pension architecture will not be reformed through direct confrontation with the OROP commitment. Any viable reform path must work within the existing entitlement structure while changing the fiscal architecture surrounding it.

5.2 The Asymmetry Between Committed and Discretionary Expenditure

A structural asymmetry sits at the heart of the crowding-out problem. Pension entitlements are legally enforceable, have been upheld in multiple Supreme Court rulings, and are reinforced by pay commission recommendations that carry quasi-statutory force. Capital outlay is discretionary in the sense that no platform acquisition programme has an equivalent legal standing. A programme can be delayed, deferred, or cancelled without breach of any legal obligation, which is precisely what has happened repeatedly across the period studied.

This asymmetry means that standard fiscal management tools operate asymmetrically within the defence budget. A Finance Ministry circular instructing ministries to compress expenditure by a given percentage applies with full force to the capital head and with near-zero practical force to the pension head. A government that genuinely sought to shift fiscal resources from pensions to capital could not achieve this through normal expenditure management instructions; it would require legislative or judicial action of a kind that is politically implausible given the electoral dynamics described above.

5.3 The National Pension System Constraint

The most frequently proposed structural remedy for the pension crowding-out problem is the extension of the National Pension System to military personnel. The NPS is a defined-contribution architecture under which the government's pension liability is fixed at the point of annual contribution rather than being open-ended through the retiree's lifetime under the defined-benefit model. Extension of NPS to the military would, over the medium to long run, eliminate the ratchet mechanism that drives the OROP-related expenditure steps.

The practical obstacles are substantial. Military personnel were explicitly excluded from NPS implementation in 2004 on the grounds of the unique risks of military service, including early death, disability, and shorter service terms that reduce the ability to accumulate an adequate corpus. Extending NPS to the military would require significantly enhanced government contribution rates and additional risk coverage arrangements to compensate for these features. It would also require managing a lengthy transition during which the government would simultaneously fund legacy defined-benefit obligations for existing and recent retirees and make NPS contributions for current serving personnel, creating a double financial burden for at least two to three decades.

More fundamentally, as Behera and Kaushal noted at MP-IDSA, the political feasibility of introducing NPS for the military is extremely low given the context in which OROP was implemented. Both major parties having claimed credit for OROP, offering NPS as the replacement retirement architecture would be widely interpreted as a reduction of veterans' entitlements. Any government that attempted it would need to demonstrate that the net present value of NPS benefits exceeded those of OROP for the affected cohorts. Given the defined-benefit nature of OROP and the contribution-based nature of NPS, that demonstration would be technically complex, politically contested, and vulnerable to adverse judicial review.

6. Agnipath: An Intertemporal Shift, Not a Resolution

The Agnipath scheme, announced on 14 June 2022, is the most significant institutional response to the pension-modernisation trade-off since the problem was first formally documented. It deserves careful analysis on its fiscal mechanics, its transition timeline, and the limits of what it can accomplish within the planning horizon of any realistic policy framework.

6.1 The Fiscal Mechanism

Agnipath recruits personnel below officer rank for a fixed four-year term. Approximately 75 percent of each cohort exits at the end of the term with a Seva Nidhi package of Rs.11.71 lakh, jointly contributed by the individual and the government, but without pension entitlement or ex-serviceman status. The remaining 25 percent are selected for permanent service, where they serve a minimum additional 15 years and accrue pension entitlement under the existing defined-benefit architecture.

The fiscal rationale rests on the Department of Military Affairs' calculation that each Agniveer from the exit cohort represents a prospective lifetime saving of approximately Rs.11.5 crore compared to a regular jawan who completes 17 years of service and retires with a full pension. Applied to cohorts of 46,000 or more recruits annually, the cumulative lifetime saving is substantial in absolute terms. The Standing Committee on Defence's

2022 projection that the pension bill would reach Rs.2.5 lakh crore by 2025 absent reform made the scale of the untreated problem equally clear.

6.2 The Intertemporal Shift and Why It Matters

Agnipath shifts the pension burden intertemporally rather than eliminating it. The scheme defers fiscal relief while simultaneously introducing near-term operational adjustment costs. This intertemporal framing is essential for any honest assessment of what the scheme can deliver within the planning horizons that matter for force modernisation.

The transition arithmetic is straightforward but its implications are frequently understated. Personnel recruited under the pre-Agnipath system before June 2022 are still serving and will continue to retire and draw pension under the existing defined-benefit architecture throughout the 2020s and into the early 2040s. The Agnipath exit cohorts begin leaving service in 2026, but the 25 percent retained cohort will not begin drawing pension until approximately 2043. The legacy liability from pre-Agnipath cohorts will not be substantially exhausted until the 2050s. The ORF analysis of February 2025 concluded that the fiscal dividend from Agnipath would not materialise until well into the 2030s; given the scale of the legacy liability, a more conservative estimate would place meaningful fiscal impact in the early 2040s.

This means that for the entire period covered by any credible budget forecast, through the 16th Finance Commission period of 2026-31 at minimum, the pension budget will continue to be driven primarily by pre-Agnipath liabilities. The scheme represents a genuine structural reform of the future pension accrual rate. It does not alter the existing stock of defined-benefit obligations by a single rupee.

6.3 Operational Adjustment Costs

The intertemporal relief of Agnipath comes with near-term operational adjustment costs that a full fiscal accounting must incorporate. A force in which 75 percent of PBOR serve for only four years has a different institutional memory profile, a different unit cohesion dynamic, and a different balance between training investment and operational service than one recruited primarily under longer-tenure arrangements. Whether the operationally relevant capability produced per rupee of personnel expenditure is higher or lower under Agnipath than under the pre-existing system is a genuinely contested empirical question. The adverse performance of inadequately trained conscripts in several recent conflicts -- notably the Russian experience in Ukraine -- illustrates the operational risk dimension that any mechanistic fiscal analysis must acknowledge.

This paper does not adjudicate this operational debate. The relevant analytical point is that Agnipath's headline fiscal rationale assumes that the reduction in pension accrual translates directly to additional modernisation resources. If the scheme imposes training cost increases, readiness gaps, or morale effects that require compensating expenditure in the stores or operations heads, the net benefit to the capital budget will be proportionally reduced. A complete fiscal assessment of Agnipath requires tracking all three channels of expenditure -- pension, stores, and capital -- simultaneously, not only the pension head in isolation.

Table 3: Agnipath Fiscal Parameters vs. Traditional Recruitment

Parameter	Traditional PBOR	Agnipath (75% exit cohort)	Agnipath (25% retained)
Service term	17+ years	4 years	4 + minimum 15 years
Pension entitlement	Defined-benefit, lifelong	None	Defined-benefit (post permanent service)
Exit package	Pension + gratuity	Rs.11.71 lakh Seva Nidhi	Regular pension on retirement

Parameter	Traditional PBOR	Agnipath (75% exit cohort)	Agnipath (25% retained)
Prospective lifetime saving vs. traditional (DMA estimate)	Baseline	Rs.11.5 crore per Agniveer	Lower saving relative to exit cohort
Fiscal dividend timeline	Baseline (ongoing)	2030s onwards (ORF, 2025 estimate)	2040s and beyond
Near-term pension budget impact	N/A	Negligible (legacy liability dominates to 2030s)	Minimal near-term; compounds from 2040s

Sources: DMA prospective lifetime saving calculation (cited in ORF Commentary, February 2025); Standing Committee on Defence 2022 projection; Agnipath scheme documentation, Ministry of Defence, June 2022; ORF Commentary, 'Looking Beyond Budget to Fund Defence', February 2025.

7. Comparative Perspectives: India in International Context

Cross-country comparisons in defence budget analysis carry significant interpretive risks. Accounting frameworks differ across nations; the treatment of pensions as part of the defence budget versus a separate social security entitlement varies by country; and force structure, strategic requirements, and industrial base conditions differ in ways that affect the optimal capital-to-personnel cost ratio. The comparisons below are therefore indicative rather than directly prescriptive. They are offered to establish the degree of India's structural divergence from comparable military powers and to identify which features of comparable systems have achieved rebalancing.

7.1 United States: Blended Retirement and the Political Feasibility Question

The US military pension system transitioned to the Blended Retirement System for personnel entering service from 2018. The pre-BRS system had a structural anomaly: personnel who left before the 20-year vesting threshold received nothing, meaning approximately 83 percent of those who served received no retirement benefit. The BRS replaced the cliff-vesting structure with a hybrid combining a reduced defined-benefit component with a Thrift Savings Plan (defined contribution) for all personnel. The transition required roughly a decade of design work and political advocacy but was eventually implemented without triggering the mass attrition or morale effects that critics predicted.

The US experience is relevant to India's situation in two specific respects. It demonstrates that a defined-benefit to hybrid transition in the military context is both technically and politically feasible, under the right conditions. It also reveals what those conditions are: there was a clear constituency for reform within the serving military, namely the 83 percent who would gain some retirement benefit under BRS that they would have received nothing under the pre-BRS cliff-vesting structure. India's OROP-based system has no equivalent internal reform constituency. The beneficiaries of the existing defined-benefit structure include all retirees and all serving personnel approaching retirement. There is no large internal group that would clearly benefit from a transition to a hybrid system. This structural difference, more than any technical consideration, explains why the US BRS path is not directly transferable to India.

7.2 China: Manpower Reduction as the Primary Rebalancing Tool

China's approach to the personnel cost burden in the defence budget has been structural downsizing rather than pension system redesign. The People's Liberation Army reduced its total manpower from approximately 2.3 million to approximately 2.0 million between 2015 and 2017, a reduction of 300,000 personnel concentrated in non-combat civilian and support roles. Fewer serving personnel means fewer future

pensioners, and the PLA's capital expenditure share has been estimated at approximately 40 percent or above of the defence budget, substantially higher than India's.

India's equivalent manpower reduction path is constrained by its two-front security environment. The Line of Actual Control with China and the Line of Control with Pakistan both require substantial infantry presence in high-altitude terrain where technological substitution is limited in the near term. A significant reduction in the Indian Army's sanctioned strength of 1.2 million would require either a strategic reassessment of the threat environment or substantial capital investment in the technology systems that would substitute for reduced manpower. The manpower reduction path is available in principle; it requires precisely the capital expenditure that the pension-driven crowding-out has constrained.

7.3 France: Career Architecture Reform Over System Redesign

France's transition to a fully professional force in 1996, abolishing conscription, was accompanied by deliberate career incentive restructuring that reduced average service lengths and created a more dynamic inflow-outflow balance. French military personnel have shorter average service terms, earlier and more structured lateral transitions to the civil service and private sector, and a pension architecture that vests later but is more portable than India's current system. The result is a manpower cost profile that is more predictable and less steep in its pension trajectory than India's defined-benefit OROP structure.

The French example illustrates the general principle that career architecture reform -- changing the structure of service terms and exit pathways -- can achieve pension cost rebalancing without requiring a direct confrontation with existing pension entitlements. Agnipath is, in design intent, an attempt to apply this principle to the Indian context. Its implementation shortcomings are well-documented: the absence of a statutory CAPF absorption quota, the lack of portable pension credits for the exit cohort, and the limited reabsorption infrastructure. These are design refinements that do not invalidate the underlying fiscal logic; they represent the unfinished work of implementation.

Table 4: Comparative Manpower-Pension Profiles (Indicative)

Feature	India	United States	China	France
Pension system type	Defined-benefit + OROP ratchet	Hybrid DB+DC (BRS, from 2018)	State-managed; not separately disclosed	Defined-benefit; shorter vesting
Pensions as % of defence budget (approx.)	21 to 24% (2020s avg.)	8 to 10% (approx.)	Not disclosed at this level	12 to 14% (approx.)
Capital % of defence budget (approx.)	24 to 28% (2020s)	40 to 45% (approx.)	40%+ (estimated)	30 to 35% (approx.)
Recent structural reform	Agnipath (2022)	Blended Retirement System (2018)	300,000 personnel reduction (2015-17)	Conscription abolition (1996)
Fiscal dividend timeline	2030s to 2040s	2028 to 2035	Near-term (reduction completed)	Near-term (realised by early 2000s)

Note: Cross-country comparisons are indicative due to differences in accounting frameworks, disclosure standards, and force structures. US and France percentage ranges are approximations from open-source analyses. China's internal budget composition is not publicly disclosed at the level of pension-capital disaggregation; the capital percentage is estimated from CSIS China Power Project analyses and should be treated as approximate. Sources: SIPRI Military Expenditure Database; US DoD Blended Retirement System documentation; CSIS China Power Project; French Ministry of Armed Forces annual reports; MP-IDSA India budget series.

8. Architectural Obstacles to Structural Resolution

8.1 The Pension-as-Separate-Head Problem

The defence pension budget is allocated under Demand No. 22 (Defence Pensions), which is a separate demand for grants from the defence services budget (Demands 20 and 21). This administrative separation creates a specific fiscal distortion. When the Finance Ministry engages in expenditure management discussions with the Ministry of Defence, Demand No. 22 is effectively treated as a committed, non-negotiable liability. Expenditure compression instructions fall on Demands 20-21, the operational defence budget. The pension head grows unimpeded while the operational budget bears the adjustment.

The consequence is that the operative defence budget -- the one that funds salaries, stores, and capital -- is being squeezed by a liability that is administratively classified elsewhere. Finance Ministry officials reviewing the 'defence budget' as a single envelope are making allocation decisions that implicitly treat pension growth as a first charge on the operational envelope without explicitly declaring it. A budget architecture that made this priority explicit -- treating pension expenditure as a separate sovereign welfare obligation funded from general revenues rather than as a component of the operational defence budget -- would force a different set of allocation decisions. Policymakers would be required to decide explicitly what proportion of the residual operational envelope goes to capital, rather than arriving at that figure as the arithmetic remainder after pension growth has been accommodated.

8.2 The Non-Lapsable Fund That Was Never Constituted

The 15th Finance Commission recommended the Modernisation Fund for Defence and Internal Security (MFDIS) as a dedicated non-lapsable fund, to be capitalised through proceeds from defence land monetisation, DPSU disinvestment, and other non-tax receipts. The Ministry of Finance rejected the recommendation on grounds of parliamentary practice and fiscal discipline, noting that a standing non-lapsable appropriation would impair parliamentary oversight of defence expenditure.

The rejection was not without procedural basis. Parliamentary sanction of a non-lapsable fund does raise genuine questions about the annual appropriation process. But the Finance Ministry's response treated the specific governance design of the proposed MFDIS as constitutionally necessary rather than as one of several possible structures. A non-lapsable fund subject to annual Parliamentary drawdown approval -- under which Parliament votes the fund balance annually and approves specific drawdowns against designated acquisition programmes -- would maintain parliamentary oversight while providing the multi-year capital commitment horizon that major acquisitions require. The Ministry of Finance's objection was to a standing appropriation; a structure requiring annual Parliamentary approval of drawdowns would address that objection without abandoning the non-lapsable modernisation fund principle. The MFDIS was rejected rather than redesigned, and the modernisation budget was left fully exposed to the annual pension squeeze with no structural buffer.

8.3 The 60:40 Revenue-to-Capital Target That Has Not Been Approached

The Standing Committee on Defence recommended an ideal revenue-to-capital ratio of 60:40 in 2021. The Indian Army's actual ratio in that year was approximately 80:20. The recommendation has been reiterated across multiple successive Standing Committee reports without generating any movement in the actual ratio. This is not a case of a target being marginally missed in certain years. It is a case of a target being consistently and substantially missed across an entire decade, with no institutional mechanism in the budget architecture that converts the recommendation into a binding allocation commitment.

The Standing Committee can recommend. It cannot compel. The Ministry of Defence can acknowledge the recommendation, as it routinely does, and then represent a request for capital funding that is scaled back by the Finance Ministry under general fiscal consolidation pressure. The Finance Ministry can note the Standing Committee recommendation in its correspondence while allocating less than what would be required to move the ratio. No actor in this chain has both the authority and the incentive to enforce the target. The ratio has not moved because the institutional design of the budget process does not create consequences for missing it.

9. A Sequenced Reform Architecture

The analysis in this paper points toward a specific sequenced reform architecture. The sequencing matters: the foundational reform creates the institutional precondition without which subsequent reforms cannot deliver their full effect.

9.1 Foundation: Structural Separation of the Pension Demand

The pension demand should be formally separated from the operational defence budget at the parliamentary appropriation level and treated as a sovereign welfare obligation funded from general revenues. This is not merely a presentation change. It means that Finance Ministry expenditure management decisions about 'the defence budget' would not affect pension payments, and that defence capital allocations would be calculated against an operational envelope that does not include pension commitments. The political economy implication is that Ministers and Parliament would be required to make explicit decisions about the pension-capital trade-off rather than allowing it to be resolved by administrative inertia in the direction of pension protection.

This reform requires no legislative amendment to the OROP structure. It does not threaten any existing pension entitlement. It changes only the administrative classification of how pension expenditure is presented and managed in relation to the operational defence budget. Its fiscal impact would be immediate: once the pension demand is outside the operational envelope, the capital share of the operational envelope would rise in a single budget cycle.

9.2 Multi-Year Capital Commitment Authority

The second tier of the reform architecture is the constitution of a Modernisation Fund for Defence and Internal Security with a governance structure that addresses the Ministry of Finance's parliamentary control concerns. The fund should be capitalised through a specific annual appropriation voted by Parliament, with drawdowns subject to annual Parliamentary approval tied to specific acquisition programmes rather than authorised as a standing appropriation. This structure maintains Parliamentary oversight while providing the multi-year capital commitment horizon that major acquisitions require. The Ministry of Finance's objection to the 15th Finance Commission's MFDIS design was to a standing, automatically available fund. An annually voted drawdown structure addresses that objection.

9.3 Agnipath Implementation Strengthening

The Agnipath scheme is directionally correct and should be retained, but its implementation requires three specific enhancements to deliver its intended fiscal and operational outcomes. First, a statutory quota for Agniveer absorption into the CAPF at the end of the four-year term -- not a preference or aspiration but a legal obligation with enforcement -- would address the civilian reabsorption problem that is the primary source of political opposition to the scheme. Second, portable NPS enrollment for all Agniveers, with accounts that transfer to CAPF or other public sector employers at exit, would create a fiscal bridge that improves exit-cohort welfare and reduces the political criticism of the scheme as abandoning veterans. Third, an independent operational review at the five-year mark (approximately 2027) with Parliamentary Standing Committee oversight should assess whether the 25 percent retention rate is operationally adequate, with provision to adjust it upward to 30 to 35 percent if unit cohesion and readiness evidence suggests it is necessary.

9.4 The 16th Finance Commission as an Institutional Opportunity

The 16th Finance Commission, covering 2026-31, represents the next formal opportunity for a structured reassessment of the pension-modernisation architecture. Its terms of reference, which are determined by the Government of India, should include a specific mandate to examine the interaction between defence

pension liabilities and the capital modernisation budget, and to provide recommendations for a revised funding architecture. A 16th Finance Commission recommendation incorporating the governance redesign of the MFDIS described above, together with a specific capital-share target tied to a monitoring mechanism, would carry substantially more institutional weight than a Standing Committee recommendation that lacks enforcement consequence.

10. Conclusion

This paper has argued, with data spanning two decades, that India's defence pension growth and its modernisation constraints are structurally linked through intra-budgetary fiscal crowding-out rather than being independent challenges. The Pension Displacement Index, constructed as an indicative measure of the directional co-movement between pension expenditure growth and capital outlay within the defence envelope, suggests that in the acute phase of OROP-driven displacement between 2011-12 and 2020-21, a substantial proportion of incremental pension expenditure corresponded with compression in capital and stores allocations. The combined capital-and-stores share of the MoD budget contracted by 11 percentage points over that decade, a result consistent with near-complete intra-budgetary substitution at the margin.

The crowding-out is structurally self-reinforcing for reasons that this paper has traced through both its fiscal architecture and its political economy. The pension ratchet created by OROP means that expenditure grows automatically with each five-year review cycle, without requiring any annual discretionary decision. The electoral lock-in created by bi-partisan OROP commitments means that no government can reduce the ratchet's step-size without severe political cost. The administrative classification of the pension demand within the operational defence budget envelope means that pension growth is accommodated as a first charge, leaving capital as the residual. And the absence of a non-lapsable modernisation fund means that capital savings from one year cannot be carried forward to fund multi-year acquisition programmes.

The Agnipath scheme is a genuine structural reform of the future pension accrual rate, but it shifts the pension burden intertemporally rather than eliminating it. Its fiscal dividend will not materialise until the 2030s at the earliest, and its near-term operational adjustment costs must be incorporated into any complete fiscal assessment. The improvement in capital shares visible in FY 2024-25 and FY 2025-26 is real but largely crisis-driven; the next OROP review cycle in 2029 will test whether the architectural improvements are durable without strategic emergency cover.

Three reforms are necessary rather than sufficient. Structural separation of the pension demand from the operational defence budget envelope would force explicit political decisions about the capital-pension trade-off. Constitution of the MFDIS with annual drawdown oversight would provide the multi-year capital commitment horizon that major acquisitions require. Strengthened Agnipath implementation with statutory CAPF absorption quotas and portable pension credits would accelerate the long-run fiscal dividend while reducing political opposition. None of these reforms requires a change to existing pension entitlements. All of them require the Ministry of Finance to accept a governance structure that partially insulates defence capital allocations from annual expenditure management pressure.

Absent structural reform of this architecture, the pension-modernisation trade-off will remain the central constraint on India's transition to a technology-intensive force structure. India will continue to buy more capability on paper than it can induct in reality -- not because the resources are absent, but because the institutional design ensures they arrive at the procurement table after everything else has already been paid.

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